

Mikhail Polykovskii

Lead Python Developer | Tech Lead | Staff Engineer

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[LinkedIn](#) | [GitLab](#) | [Telegram](#)

Summary

Python backend engineer with 17+ years of experience building and launching production systems from scratch. Delivered multiple production projects in the last 3 years: AI-powered content pipelines, customer data enrichment platforms, SSO integrations. Led teams of up to 10 engineers — known for clear task structuring, fast problem-solving, and creating a collaborative environment. AWS Certified Solutions Architect with deep expertise in microservices, high-load systems, and full SDLC ownership. Fluent in modern AI coding tools (Claude Code, Codex) to accelerate development.

Skills

Languages & Frameworks:	Python, Django, FastAPI, Flask, Celery, SQLAlchemy, Alembic, Pydantic
Testing:	Pytest, TDD, unittest
Databases:	PostgreSQL, ClickHouse, Snowflake, DynamoDB, Redis, Elasticsearch, MongoDB
Infrastructure:	Docker, Kubernetes, Helm, AWS (EC2, S3, Lambda, RDS, ECS, API Gateway, CloudWatch, IAM), GCP
Queues & Messaging:	Kafka, Redis, AWS SQS
Architecture:	API Gateway, Circuit Breaker, Event Sourcing, CQRS, Saga pattern, Service Mesh (Istio)
Batch Processing:	Airflow
Observability:	Prometheus, Grafana, Sentry
CI/CD:	GitHub Actions, GitLab CI, ArgoCD, Jenkins
Practices:	Agile, Scrum, TDD, DDD, CI/CD
AI Tools:	Claude Code, Codex, OpenAI API

Experience

Lead Python Developer | US based software company 04/2023 - Present

Web development company

FastAPI, PostgreSQL, Kafka, Kubernetes, GCP, Snowflake, OpenAI API, AWS Cognito, AWS Lambda

AI-Powered Sports Content Pipeline

- Built end-to-end microservices pipeline ingesting rich sports event data (match statistics, player actions, play-by-play events) from providers via polling
- Solved the core challenge of selecting relevant data subsets from high-volume inputs to construct effective LLM prompts; generated engaging articles via ChatGPT (OpenAI API)
- Generated 50,000+ articles at launch; system continues producing content weekly

SSO Integration (AWS Cognito)

- Implemented seamless SSO login for an existing AWS Lambda-based system
- Integrated Microsoft, Google, and Apple as identity providers using OAuth 2.0, OIDC,

and SAML

- Delivered zero-downtime migration with no disruption to existing users

Customer Data Enrichment Platform

- Architected and built FastAPI backend from scratch for a platform ingesting customer sales, donation, and visit data into Snowflake (hundreds of millions of rows)
- Enforced security model: role-based access control, row-level permissions, user scopes, and audit logging
- Collaborated closely with React frontend team; drove architectural decisions and kept tech debt under control
- Spec Driven Development with Claude Code

Lead Python Developer | FoodRocket

11/2022 - 03/2023

Food delivery startup, US market: Chicago, San Francisco

FastAPI, PostgreSQL, Kafka, Kubernetes, GCP

- Developed courier management microservice from scratch for dark store delivery operations
- Shipped new features while managing accumulated tech debt in a fast-moving startup codebase
- Team: 4 backend engineers, 2 QA, 1 PM; coordinated with separate mobile and ops teams

Lead Software Engineer | Quantori LLC

01/2022 - 11/2022

Life science technology company

Django, PostgreSQL, Redis, Kubernetes, AWS, Apache Airflow, CWL

- Led team of 10 engineers building a high-performance molecular search tool supporting multiple molecular encoding standards and search methods
- Established Agile development processes (sprints, code review, release cadence), increasing release frequency
- Added automated error monitoring across all environments

Research Publication

- Co-authored poster with Harvard University FAS RC: "*Reproducible Data Engineering Pipelines using Common Workflow Language and Apache Airflow*" — [f1000research poster](#)

Backend Team Lead | SBDA Group

03/2020 - 12/2021

ML-powered retail analytics

Django, DRF, Airflow, Hadoop, ClickHouse, PostgreSQL

- Led team of 7 building demand forecasting visualization interface for X5 Retail Group (one of Russia's largest retail chains)
- Solved high-volume data access bottleneck: designed ETL pipeline migrating prediction data from Hadoop to ClickHouse, enabling fast UI query response
- Engineered workaround for ClickHouse immutability to support manual forecast corrections by retail managers
- Owned CI/CD setup and all technical decision-making across the team

Senior Software Engineer | Iponweb

04/2019 - 02/2020

Global advertising technology platform

Django, DRF, RTB APIs

- Built back-office interfaces for large-scale ad inventory sellers with complex, highly interdependent data models
- Integrated RTB (Real-Time Bidding) ad display system APIs

- Improved performance of company-wide internal Django framework

Python Backend Developer | Yandex

04/2007 - 07/2018

#1 tech company in Russia, 100M+ monthly users

Python, Django, PostgreSQL

Internal Access Control System

- Built fine-grained permission management system handling highly complex and variable access rules across internal services

Yandex.Taxi (early-stage team)

- Backend developer on the original Python-based Yandex.Taxi launch team; delivered features in a startup-within-enterprise environment

Olympic & FIFA World Cup Sports Services

- Delivered sports event services under immovable hard deadlines (Olympic schedule)
- Designed high-load architecture decoupling frontend from backend: backend pre-computed and cached pages/data, frontend served from cache — ensuring stability under major traffic spikes

Yandex.Schedules (greenfield launch)

- Launched train, flight, and suburban transport scheduling service from scratch — one of Russia's first such services
- Adopted Django in its early days; later extended platform to include ticket sales

Education & Certifications

AWS Certified Solutions Architect - Associate

Master's Degree in Engineering

Ural State University | Mathematics and Computer Science | 1996 - 2001

Open Source

Contributed to [NSAPH-Data-Platform](#) (Harvard University) during Quantori engagement — public commits on CWL/Airflow data pipeline tooling

Languages

English B2 ([EFSET Certificate](#)) | Russian C2 (Native)

Additional Experience

MySQL / MariaDB, SQLite, Memcached, AWS SNS, aiohttp, asyncio, HTTPX, WebSockets, OpenAPI / Swagger, DigitalOcean, Jaeger, OpenTelemetry, ELK Stack (Elasticsearch + Logstash + Kibana), Docker Compose, Nginx, JWT, Keycloak, Certbot / Let's Encrypt, Yandex Tank, Postman, Travis CI, Makefile, Feature Flags, Blue/Green deployment, Canary deployment, Celery Beat, Git (advanced), Linux / bash scripting

Can Ramp Up Quickly If Needed

Tornado, Twisted, Pandas, GraphQL, gRPC, JSON-RPC, SOAP, Azure, Heroku, Hadoop, Terraform, Ansible, HashiCorp Vault, HAProxy, Bitbucket Pipelines, Chaos Engineering, etcd, Protocol Buffers (protobuf)